Durability of GFRP Reinforcing Bars Embedded in Moist Concrete

Mathieu Robert[1]; Patrice Cousin[2]; and Brahim Benmokrane[3]

Abstract: This paper presents mechanical, microstructural, and physical characterization of glass fiber-reinforced polymer (GFRP) bars exposed to concrete environment. GFRP bars were embedded in concrete and exposed to tap water at 23, 40, and 50°C to accelerate the effect of the concrete environment. The measured tensile strengths of the bars before and after exposure were considered as a measure of the durability performance of the specimens and were used for long-term properties prediction based on the Arrhenius theory. In addition, Fourier transform infrared spectroscopy, differential scanning calorimetry, and scanning electron microscopy were used to characterize the aging effect on the GFRP reinforcing bars. The results showed that the durability of mortar-wrapped GFRP bars and exposed to tap water was less affected by accelerated aging than the bars exposed to simulated pore-water solution. These results confirmed that the concerns about the durability of GFRP bars in concrete, based on simulated laboratory studies in alkaline solutions, do not properly correspond to the actual service life in concrete environments.

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CE Database subject headings: Durability; Fiber reinforced polymers; Moisture; Aging; Alkalinity; Predictions; Mechanical properties; Microstructures.

Introduction

Fiber-reinforced polymers (FRP) have been widely used in aeronautical and chemical engineering for decades. FRP materials are also increasingly being used for engineering applications. However, the low cost-to-performance advantage of glass fiber reinforced polymers (GFRP) is the driving force behind its worldwide use and acceptance. GFRP materials are attributed to being of high strength, light weight, noncorrosive, and nonconductive. Unfortunately, in some special conditions, such as in a high alkalinity environment, the long-term performance of the GFRP is still an unresolved question. Strength of the glass fibers and resin matrix, the two constituents of the GFRP materials, can decrease when subjected to a wet alkaline environment.

Several research works were carried out to investigate the durability of GFRP materials under different environmental conditions that are anticipated under actual service conditions. Sen et al. (1999) investigated the durability of S-2 glass/epoxy prestressed beams exposed to wet/dry cycles in a 15% salt solution and found that GFRP bars lost their effectiveness within 3–9 months of exposure. S-2 glass has many of the features of glass fiber with increased strength and temperature resistance. Porter et al. (1997) exposed three different types of E-glass FRP bars (manufactured using an isophthalic polyester resin) to high alkaline solution and a maximum temperature of 60°C for a period of 2–3 months. Their test results indicated that the accelerated aging severely reduced the ultimate tensile strength and the maximum strain capacity of the GFRP bars. Dejke (1999) reported that glass fibers are known to degrade in the presence of water, and that moisture can decrease the glass transition temperature (T_g) of the resin and act as a plasticizer, potentially having significant effect on flexural strength. The reaction of FRP composite with alkali of concrete is one of the major durability concerns for design engineers. Typically, concrete environment has high alkalinity, which depends on the design mixture of the concrete, and the type of cement used (Diamond 1981; Taylor 1987). This alkaline environment damages glass fibers through loss in toughness, strength, and embrittlement. Recently, Chen et al. (2007) studied the durability of bare FRP bars and bars embedded in concrete immersed in different solutions at different temperatures and time of exposure. The writers found that GFRP bare bars and embedded in concrete bars showed a significant strength loss when exposed to simulated environments, especially for solutions at high temperature of 60°C (Chen et al. 2007).

Glass fibers are damaged due to the combination of two processes: (1) chemical attack on the glass fibers by the alkaline cement environment; and (2) concentration and growth of hydration products between individual filaments (Murphy et al. 1999). The embrittlement of fibers is due to the nucleation of calcium hydroxide on the fiber surface. The hydroxylation can cause fiber surface pitting and roughness, which act as flaws severely reducing fiber properties in the presence of moisture. In addition, calcium, sodium, and potassium hydroxides found in the concrete pore solution are aggressive toward glass fibers (Benmokrane et al. 2002, Nkurunziza 2004). Therefore, the degradation of glass

[1]Ph.D. Candidate, Dept. of Civil Engineering, Univ. of Sherbrooke, Sherbrooke PQ, Canada J1K 2R1. E-mail: mathieu.robert2@usherbrooke.ca

[2]Research Associate, Dept. of Civil Engineering, Univ. of Sherbrooke, Sherbrooke PQ, Canada J1K 2R1. E-mail: patrice.cousin@usherbrooke.ca

[3]NSERC Research Chair Professor in Innovative FRP Composite Materials for Infrastructures, Dept. of Civil Engineering, Univ. of Sherbrooke, Sherbrooke PQ, Canada J1K 2R1 (corresponding author). E-mail: Brahim.Benmokrane@USherbrooke.ca

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fibers not only depends on the high pH level, but also on the combination of alkali salts, pH, and moisture.

Aqueous solutions with high pH are known to degrade the tensile strength of GFRP bars (Chen et al. 2006), although particular results strongly vary according to differences in test methods. Higher temperature and long exposure time exasperate the problem. Most of the data were generated using temperatures as low as slightly below the freezing point and as high as a few degrees below the glass transition temperature of the resin. Tensile strength reductions in GFRP bars ranging from 0 to 75% of initial values have been reported in the literature cited (Katsuki and Uomoto 1995; Dejke 1999; Chen et al. 2006). Hence, it is clear that FRP material components and test methods have an important influence on the conclusions regarding the durability of FRP bars in concrete (Mufti et al. 2007).

In order to evaluate long-term durability performance of FRP in an alkaline environment, extensive studies have been conducted to develop accelerated aging procedures and predictive models for long-term strength estimates, especially for GFRP bars (Porter et al. 1997; Dejke 2001; Bank et al. 2003; Chen et al. 2006). These models are based on an Arrhenius-type model proposed by Litherland and his colleagues (Litherland et al. 1981). Research studies on the effects of temperature on the durability of FRP bars in concrete alkaline environment indicates that an accelerated factor for each temperature difference can be defined by using Arrhenius laws. These factors differ for each product, depending on the types of fiber and resin and bar size. In addition, the factors are affected by the environmental conditions, such as surrounding solution media, temperature, pH, moisture, and freeze-thaw conditions. Predictive models based on Arrhenius laws make the implicit assumption that the elevated temperature will only increase the rate of degradation without affecting the degradation mechanism or introducing other degradation mechanisms of FRP bars. Gerritse (1998) indicated that at least three elevated temperatures are necessary to perform an accurate predication based on Arrhenius law. Moreover, the measured data should be in continuous time intervals.

Until recently, experts were not in full agreement about whether or not GFRP is stable in the alkaline environment of concrete. In a recent study (Mufti et al. 2007), nine cores were taken from each of five in-service concrete bridge structures across Canada, reinforced with GFRP bars; these structures were built during the last 6–8 years. Three cores from each bridge were given to each of three teams of material scientists and experts in durability, for microscopic and chemical analyses. An example from this study is presented in Fig. 1, which shows a micrograph of GFRP and surrounding concrete removed from an 8-year-old structure. It can be seen that the glass fibers and the GFRP/concrete interface are intact. The findings from the analyses reported by Mufti et al. (2007) have confirmed that the concerns about the durability of GFRP in alkaline concrete, based on simulated laboratory studies in alkaline solutions, are unfounded. It is mainly on the basis of this study that GFRP is now permitted by the Canadian highway bridge design code (CSA 2006) as both primary reinforcement and prestressing tendons in concrete (Mufti et al. 2007).

The current study is aimed to simulate and approach real conditions by immersing mortar-wrapped GFRP bars in tap water. The conditioning used in the present study is closer to field conditions because the FRP material is embedded in concrete which is the actual situation in the field. Indeed, the main objective of this study is to show that the traditional accelerated aging in alkaline solution is too severe and prematurely degrades the GFRP reinforcing bars, leading to limited life expectancies and too conservative predictions.

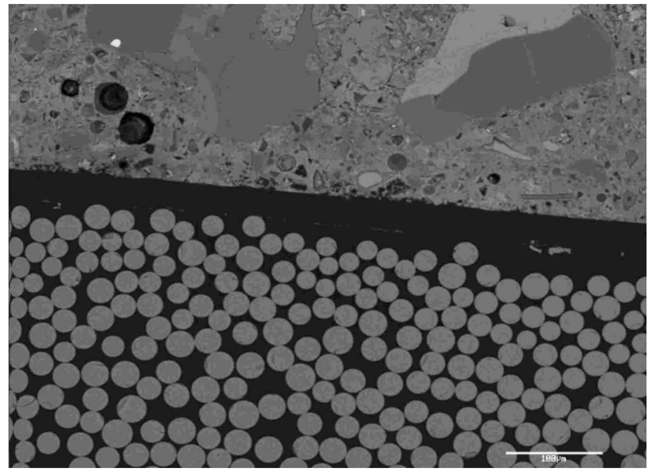

Fig. 1. Micrograph of GFRP bar/concrete interface in specimen removed from 8-year-old GFRP reinforced concrete structure (Mufti et al. 2007)

Experimental Program

Material

Sand coated glass FRP bars manufactured by a Canadian company (Pultrall Inc. 2005) were used in this study. The bars were made of continuous E glass fibers impregnated in a vinylester resin using the pultrusion process. The mass fraction of glass is 81.5% and was determined by thermogravimetric analysis according to ASTM E 1131 (ASTM 2003b) standard. Their relative density according to ASTM D 792 standard is 1.99 (ASTM 2000). The mechanical and physical properties of 12.7 mm diameter GFRP bars, as provided by the manufacturer, are summarized in Table 1. The GFRP bars used have a nominal diameter of 12.7 mm. All bars were cut into 1,440 mm lengths as specified by the ACI 440.3R-04 B2. The bars were divided into two series: (1) the unconditioned reference samples; and (2) conditioned samples (60 bars) embedded in concrete. The mortar mixture consisted of three parts of sand, one part of Type I cement according to ASTM C 150 (ASTM 2005) standard, and a water/cement ratio of 0.40 which led to a concrete pH of 12.15 measured by extraction of the interstitial solution after aging. The concrete (or mortar) was cast only in the middle third of the bars to avoid any degradation at the ends which were used as grips during the tensile tests ACI440.3R-04 B2. The concrete mold of the envelope was made of wood having a square section of 48 mm that leads to a minimum concrete cover of 18 mm. Fig. 2 shows a sketch illustrating a mortar-wrapped bar, whereas Fig. 3 shows a picture of a mortar-wrapped bar.

Test Plan

Accelerated aging of GFRP reinforcing bars embedded in concrete, used in this study, were designed to simulate an aggressive alkaline environment of saturated concrete. The embedded samples were immersed in tap water which differs from the conventional accelerated aging tests where bare bars are directly immersed in alkaline solutions simulating the aforementioned

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Table 1. Mechanical and Physical Properties of 12.7 mm Diameter GFRP Bar (as Provided by Manufacturer)

	Property	Unity	Value
Mechanical properties	Nominal tensile strength	MPa	786
	Guaranteed design tensile strength	MPa	708
	Nominal tensile modulus	GPa	46.3
	Tensile strain	%	1.70
	Poisson's ratio	—	0.26
	Nominal flexural strength	MPa	1,005
	Nominal flexural modulus	GPa	46.8
	Flexural strain	%	2.15
	Nominal compressive strength	MPa	473
	Nominal shear strength	MPa	212
Physical properties	Longitudinal coefficient of thermal expansion	$\times 10^{-6}/°C$	5.5
	Transverse coefficient of thermal expansion	$\times 10^{-6}/°C$	29.5
	Moisture absorption	%	0.38
	Glass content	% volume	60.3
		% by weight	77.9

results. The technique currently used is believed to be more representative of the real life situation. Indeed, the pH of the solution surrounding the bars is a result of the absorption of water by the concrete, thus allowing the liberation of the alkaline ions of the concrete directly in the bars environment. The aging was performed by immersing the mortar-wrapped GFRP bars in a wood container specially manufactured for the study. Fig. 4 shows the containers that were tightly closed with a polyethylene film on their inner surfaces. A polyethylene sheet was also placed on top of the wood containers to avoid excessive evaporation of water during the conditioning. Bars were spaced from each other and from the bottom of the container to allow free circulation of the solution between and around the GFRP bars. Furthermore, the water level was kept constant throughout the study to avoid a pH increase which could be due to a water level decrease and a significant increase of the concentration of the alkaline ions in the solution. The temperatures of immersion were chosen to accelerate the degradation effect of aging; however, they were not too high to produce any thermal degradation mechanisms.

The specimens were completely immersed at three different temperatures (23, 40, and 50°C) and were removed from the water after four different periods of time (60, 120, 180, and 240 days). After each period, usually six GFRP bars were removed from the water and tested under tension to compare their tensile strength retention values to those of the reference specimens. After immersion, no significant change was observed at the surface of GFRP bars.

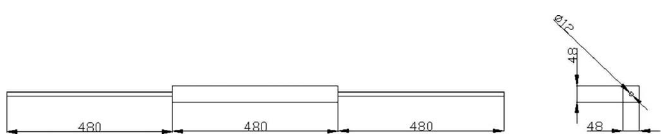

Fig. 2. Sketch illustrating typical cement mortar-wrapped GFRP bar specimen

Tensile Tests

All bars were tested under tension according to the ACI 440.3R-04 B2. Each specimen was instrumented with a linear variable differential transformer (LVDT) to capture the elongation during testing. The test was carried out using a Baldwin testing machine and the load was increased until failure. For each tensile test, the specimen was mounted on the press with the steel pipe anchors gripped by the wedges of the upper and the lower jaw of the machine. To avoid any damage to the bar, the concrete cover

Fig. 3. View of cement mortar-wrapped GFRP bar specimen

Fig. 4. Wood container built for aging of cement mortar-wrapped GFRP bar specimens

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was carefully removed from the middle third of the specimens with a hammer. The rate of loading ranged between 250 and 500 MPa/min. The applied load and bar elongation were recorded during the test using a data acquisition system monitored by a computer. Due to the brittle nature of GFRP, no yielding occurs and the stress-strain behavior was linear.

Arrhenius Relation

Eq. (1) expresses the Arrhenius relation, in terms of the degradation rate (Nelson 1990)

$$k = A\exp\left(\frac{-E_a}{RT}\right) \tag{1}$$

where k=degradation rate (L/time); A=constant relative to the material and degradation process; E_a=activation energy of the reaction; R=universal gas constant; and T=temperature (K). The primary assumption of this model is that only one dominant degradation mechanism of the material operates during the reaction and that this mechanism will not change with time and temperature during the exposure (Chen et al. 2006). Only the rate of degradation will be accelerated with the temperature increase. Eq. (1) can be transformed into

$$\frac{1}{k} = \frac{1}{A}\exp\left(\frac{E_a}{RT}\right) \tag{2}$$

$$\ln\left(\frac{1}{k}\right) = \frac{E_a}{R}\frac{1}{T} - \ln(A) \tag{3}$$

From Eq. (2), the degradation rate k can be expressed as the inverse of time needed for a material property to reach a given value (Chen et al. 2006). From Eq. (3) one can further observe that the logarithm of time needed for a material property to reach a given value is a linear function of $1/T$ with the slope of E_a/R (Chen et al. 2006). Ea and A can be easily calculated by using the slope of the regression and the point of intersection between the regression and the Y axis, respectively.

Scanning Electron Microscopy

Scanning electron microscopy (SEM) observations and image analysis were performed to observe the microstructure of specimens before and after aging. Samples observed in the SEM were the unconditioned specimens and specimens aged during 8 months in tap water at 50°C, which is a harsher aging. All specimens observed in the SEM were first cut, polished, and coated with a thin layer of gold-palladium by a vapor-deposit process. After coating the surfaces, microstructural observations were performed on a JEOL JSM-840A SEM. These observations were conducted to see the potential degradation of polymer matrix, glass fibers, or interfaces, if any.

Differential Scanning Calorimetry

Twelve–15 mg specimens from both unconditioned and aged samples were sealed in aluminum pans and analyzed in a TA Instruments DSC Q10 calorimeter equipped with a refrigerated cooling system. Analysis was conducted in modulated differential scanning calorimetry (DSC) mode. Specimens were heated from 25 to 195°C at a rate of 5°C/min. Glass transition temperature was determined for both specimens in accordance with ASTM E 1356 standard (ASTM 2003a). Two scans were performed for each specimen. The first scan is useful to determine the difference of T_g between reference and conditioned specimens. If a decrease of T_g is observed for conditioned samples, this is an indication of plasticizing effect or chemical degradation. The second scan gives information about the mechanism of degradation. A shift for higher T_g could be observed due to a post cure during the first scan. However, if after the second scan, the T_g of aged sample is in the same range as the T_g of reference sample, this is a reversible plasticizing effect due to the moisture absorption, but if the aged sample keeps a T_g lower than for the reference, this is a irreversible chemical degradation.

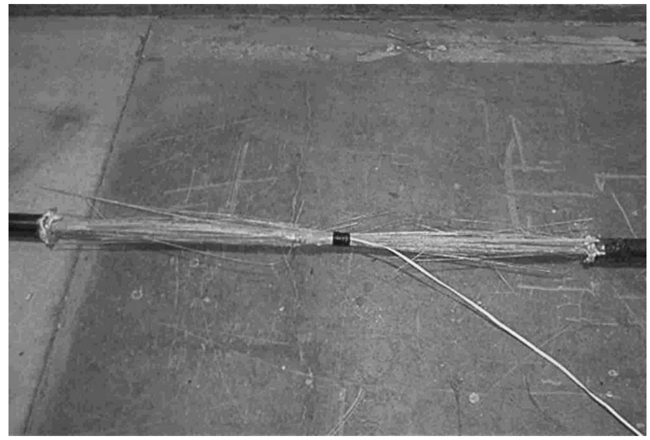

Fig. 5. Typical failure mode of conditioned GFRP bar specimens subjected to tensile test

Fourier Transform Infrared Spectroscopy

Fourier transform infrared spectroscopy (FTIR) spectra were recorded using a Nicolet Magna-550 spectrometer equipped with a attenuated total reflectance (ATR) device. Fifty scans were routinely acquired with an optical retardation of 0.25 cm to yield a resolution of 4 cm^{-1}.

Tests Results and Discussion

Tensile Strength Retention

The tensile test of unconditioned specimens showed an approximately linear behavior up to failure. Specimens failed through the rupture of fibers. The failure was accompanied by the delamination of fibers and resin, as shown in Fig. 5. A similar, but less catastrophic failure, was observed for specimens embedded in concrete and immersed in water. No chemical deposit was observed on the surface of the bars before testing. Micelli and Nanni (2004) also observed similar tensile failure modes of GFRP bars.

Table 2 shows the experimental results obtained during the tensile tests concerning the ultimate strength of aged bars tested after immersion. Fig. 6 shows the retention of the ultimate strength of aged bars according to the duration of immersion of embedded bars at various temperatures. As shown in Table 2, the tensile strength for the unconditioned bar was equal to 788 ± 54 MPa. Note that the tensile strength of bars was reduced to 665 ± 62 MPa after 240 days exposure to water at 50°C. Also, the result obtained after a conditioning of 120 days at 40°C was rejected for long-term prediction, because the mode of failure

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Table 2. Experimental Tensile Strength of Reference and Conditioned Specimens

Time of immersion (days)	Temperature (°C)	Mean tensile strength (MPa)	Coefficient of variation
0	23	788	0.069
	23	753	0.082
60	40	755	0.037
	50	767	0.038
	23	702	0.020
120	40	666	0.078
	50	720	0.032
	23	717	0.026
180	40	708	0.048
	50	711	0.025
	23	714	0.035
240	40	708	0.071
	50	665	0.093

observed during the tensile test was different than those observed by testing conditioned samples at different temperatures.

Fig. 6 shows a slight decrease of the ultimate tensile strength with immersion duration. The recorded results show that the longer the time of immersion, the larger the loss of resistance. Furthermore, it is clear that the temperature of immersion affects the loss of resistance. It can be seen that for duration of immersion of 8 months, the loss of resistance is equal to 16, 10, and 9% at 50, 40, and 23°C, respectively. This phenomenon is due to the increase of the diffusion rate of the solution and to the acceleration of the chemicals reaction of degradation with the temperature of immersion, leading to a larger absorption rate of the solution for the same time of immersion and accelerated reaction of degradation. The absorption of solution can lead to a degradation of the fibers and fiber/matrix interface, leading to a loss of the ultimate tensile. In a similar study, Wang (2005) recorded losses of resistance of 51, 25, and 32% after 300 days aging in alkaline solution (pH 12.8) of the same bars at 60, 40, and 23°C, respectively. So, for a similar period of exposure in the solutions of the same pH value, the losses of tensile strength of the GFRP bars aged in alkaline solution were more than three times larger than those of the GFRP bars aged in moist concrete.

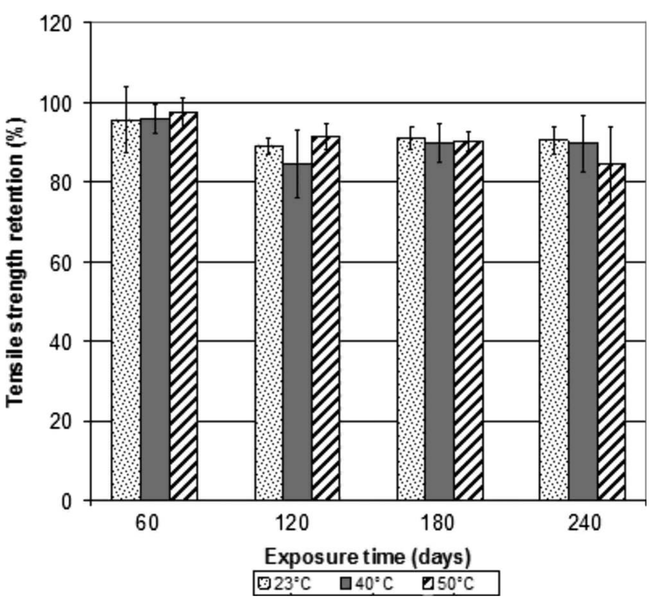


Fig. 6. Tensile strength retention of conditioned GFRP bars aged in moist concrete at 23, 40, and 50°C

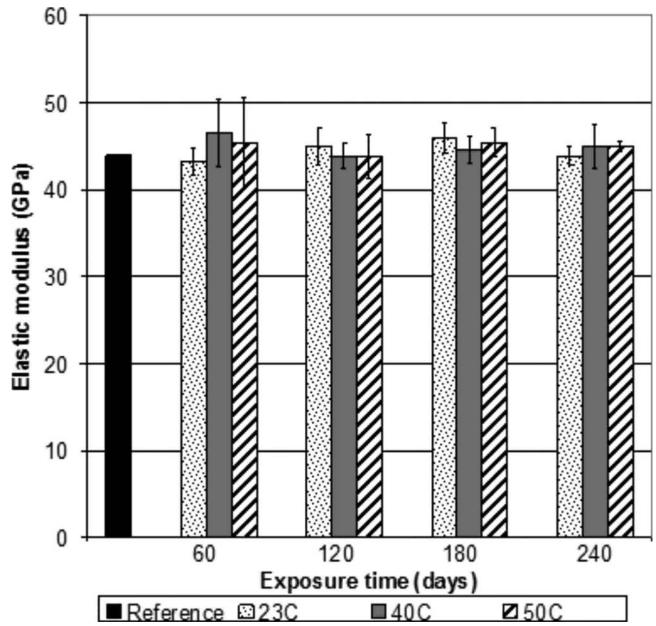


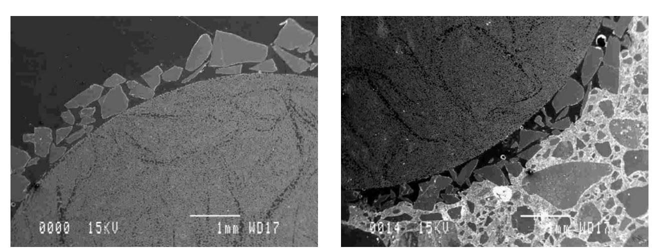
Fig. 7. Elastic modulus retention of conditioned GFRP bars aged in moist concrete at 23, 40, and 50°C

Young's Modulus

Fig. 7 shows the change in the elastic modulus of aged bars with time of immersion at various temperatures. Indeed, it can be seen from the measured results that after 240 days, the loss of elastic modulus is negligible and all aged bars are not affected by the higher temperature or the exposure to moist concrete. This result shows that elastic modulus of bars is not affected by aging in a concrete environment.

Microstructural Effects

The visual and microstructural observations showed no significant damage after 240 days of immersion in tap water at the highest temperature (50°C). The micrographs of Fig. 8 show the inter-



a) b)

Fig. 8. Micrograph (X20) of: (a) unconditioned bar; (b) mortar-wrapped FRP bar aged in water at 50°C for 240 days

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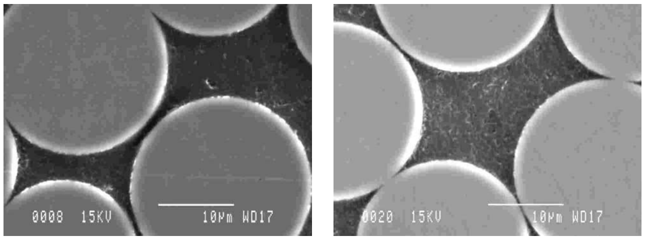


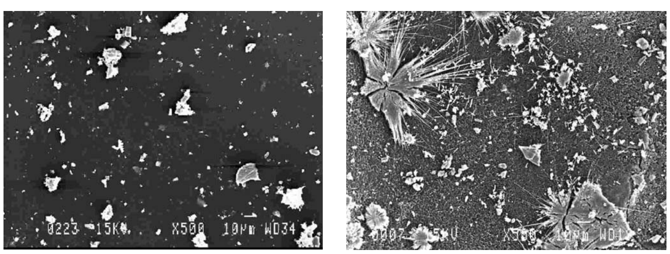
a) b)

Fig. 9. Micrographs of fiber/matrix interface (X3000) of: (a) unconditioned GFRP bar specimen; (b) conditioned GFRP bar specimen aged in moist concrete at 50°C for 240 days

face between the mixture of fibers and resin and the silica coating of the GFRP bar for an unconditioned bar and for a mortar-wrapped bar and aged in tap water for 240 days. Fig. 9 shows micrographs of the fibers/matrix interface for the same sample.

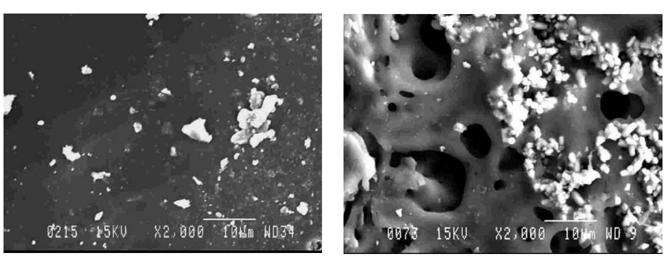
Observation of these interfaces and of the microstructure, in general, demonstrate that the conditionings of mortar-wrapped bars in water do not affect the microstructural properties of the GFRP bars; this contradicts what several researchers had noted earlier when similar GFRP bars were directly immersed into an alkaline solution (Benmokrane and Wang 2002). This phenomenon illustrates the fact that GFRP bars are not significantly affected by accelerated aging.

This observation proposes that the standard accelerated aging in alkaline solution could very severely alter the properties of aged bars compared to tap water aging which could to be a better simulation of the real service.



(a) Unconditioned GFRP bar specimen at low magnification

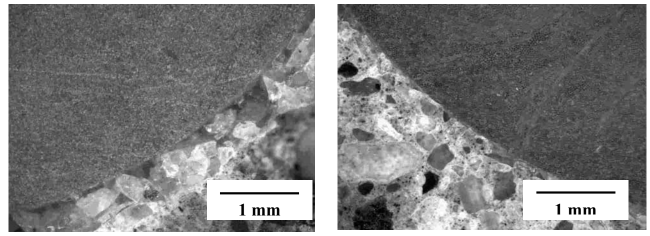
(b) Surface of GFRP bar specimen aged in moist concrete at 50°C for 240 days at low magnification



(c) Surface of GFRP bar specimen aged in moist concrete at 50°C for 240 days at high magnification

(d) GFRP bar specimen aged in alkaline solution at 60°C for 300 days at high magnification (Wang, 2005)

Fig. 10. Comparison of micrographs of external surfaces of GFRP bar specimens aged in moist concrete (present study) and in alkaline solution (Wang 2005)



(a) Unconditioned GFRP bar

(b) Conditioned GFRP bar aged in moist concrete at 50°C for 240 days

Fig. 11. Micrographs of GFRP bar/concrete interface of: (a) unconditioned bar; (b) conditioned GFRP bar aged in moist concrete at 50°C for 240 days

Micrographs of External Surfaces of GFRP Bar

The micrographs of external surfaces of GFRP bar specimens aged in moist concrete (present study) and in alkaline solution (Wang 2005) are shown in Fig. 10. It should be noted that in his durability tests Wang (2005) has used the same GFRP bar as was used in the present study. The comparison of these micrographs shows that there was no significant damage that occurred to the mortar-wrapped GFRP bar surface [Figs. 10(b and c)], whereas the damage was clear on the bar aged in an alkaline solution [Fig. 10(d)]. Moreover, the GFRP bar aged in alkaline solution shows a porous surface due to its direct contact with the solution [Fig. 10(d)].

GFRP Bar/Concrete Interface

The three main components of the bond between the GFRP reinforcing bar and concrete are: (1) the chemical adhesion; (2) the mechanical interlocks; and (3) the friction. Therefore, if there is any degradation at the resin or the fibers at the interface, it is expected that those mechanisms and the durability of GFRP reinforced structure could be affected. Fig. 11 shows the GFRP bar/concrete interface for the mortar-wrapped GFRP bar and aged in water at 50°C for 240 days. Again, it can be seen that there was no significant damage to the interface between the GFRP bar and concrete after aging in tap water.

Effects on Matrix

A FTIR analysis of unconditioned and 12.7 mm mortar-wrapped bars and aged in water at 50°C for 240 days has been conducted (Fig. 12). The most interesting region of the FTIR spectra is located between 3,300 and 3,600 cm^{-1}, which corresponds to the stretching mode of the hydroxyl groups of the vinylester resin.

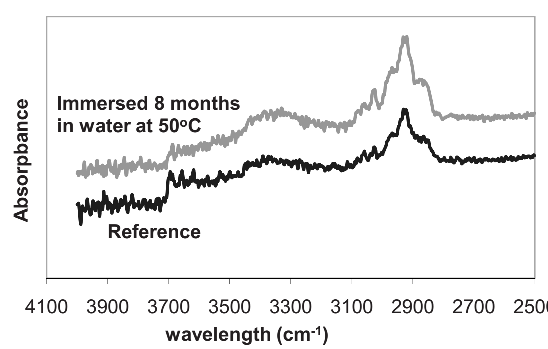


Fig. 12. FTIR spectra for unconditioned and aged samples

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Table 3. Results of Differential Scanning Calorimetry Analysis

Conditioning	Temperature (°C)	Duration (days)	T_g Run 1 (°C)	T_g Run 2 (°C)
Unconditioned	—	—	105	134
Embedded in concrete and immersed in water	50	240	104	129

When a hydrolysis reaction occurs, new hydroxyl groups are formed and the corresponding infrared band increases. Changes in the peak intensity were quantified by determining the ratio of the OH peak to the carbon–hydrogen stretching peak of the resin, which is not affected by the conditioning. The experimental ratio of the OH peak to the carbon–hydrogen stretching peak of the resin for the 12.7 mm diameter mortar-wrapped samples and immersed in water for 240 days at 50°C was 0.25 compared to 0.21 for unconditioned samples. Therefore, the hydroxyl peak does not show any significant changes. This indicates that the hydrolysis did not significantly occur in these environmental conditions. Wang (2005) obtained changes in ratio up to 30% when samples were immersed in alkaline solution at 60°C for 300 days. These results show that alkaline solutions are too harsh as compared to the immersion of the mortar-wrapped GFRP bar in water, which better simulate the real application environment.

Table 3 presents the glass transition temperature (T_g) values for the first and second heating unconditioned and aged samples. Note that for the unconditioned and aged samples, the T_g corresponding to the second heating run is higher than that of the first scan. This shift indicates that the samples were not fully cured and that a postcuring phenomenon occurred during the second heating run. However, it can be seen from the results presented in Table 3 that no significant changes of the T_g value occur after aging in water at 50°C for 240 days. This result shows that no major effect on the thermal properties of the resin, which could occur after conditioning of mortar-wrapped FRP bars, was detected by DSC.

Prediction of Long-Term Behavior

Following the procedure proposed by Bank et al. (2003), the natural logarithm of time to reach a set of levels of normalized performances versus $1/T$, expressed as the inverse of absolute temperature (1,000/K), was used to predict the service life at the mean annual temperature (6°C) in Montréal. A coefficient of determination (R^2) value close to 1 is desired. However, the ASTM procedures recommend a minimum value of 0.80 for acceptability and the obtained R^2 values are between 0.94 and 0.99. From the Arrhenius plot, the service lifetime necessary to reach the established tensile strength retention levels (PR) can be extrapolated for any temperature. Consequently, predictions are made for tensile strength retention as a function of time for an immersion at 6°C and the general relation between the PR and the predicted service life at the average temperature of 6°C are drawn (Fig. 13). The predicted time to reach the determined strength property retention level (PR) for GFRP bars embedded in moist concrete at an isotherm temperature of 6°C is approximately 1 and 210 years for a PR of 90 and 75%, respectively. The predicted service life of GFRP bars embedded in moist concrete at an isotherm temperature of 6°C to reach a PR of less than 75% can also be estimated to be infinite. These predictions show that the GFRP bars are a durable face to the concrete environment, which is supposed to be well simulated by the immersion of embedded GFRP bars in water. It is also evident that the traditional environmental aging performed through immersion in simulated concrete pore solution is much harsher than the real concrete environment. Chen et al. (2006) found that the necessary time to reach a PR of 75% for similar GFRP bars, yet immersed in an alkaline solution, was approximately 1.4 years at 20°C; the duration is very low compared to more than 27 years for mortar-wrapped bars exposed to tap water at 23°C, as shown in Fig. 13.

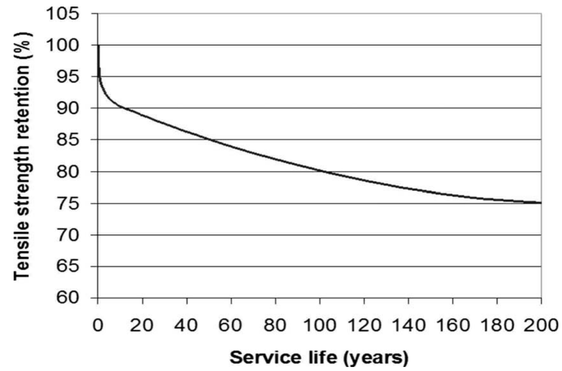

Fig. 13. General relation between PR and predicted service life at mean annual temperature of 6°C (Montreal)

Summary and Conclusions

Based on the results of this study the following conclusions may be drawn on only the product tested:

1. The aging of mortar-wrapped GFRP bars in tap water is significantly less pronounced than traditional aging in simulated pore solution. The loss in the tensile strength and the microstructural/physical effects were significantly reduced;
2. Even at high temperature (50°C), where the environment is more aggressive, the change in tensile strength is minor. For example, increasing the temperature of the solution from 40 to 50°C during 240 days results in a decrease of the tensile strength retention by 10–16% of the original tensile strength;
3. No significant microstructural changes were observed after 240 days immersion of GFRP bars embedded in concrete in tap water at 50°C. The interfaces between the bars and concrete and between the resin and the fibers do not seem to be affect by the moisture absorption and high temperatures;
4. The polymer matrix is not affected by moisture absorption and high temperatures: no changes of the glass transition temperature occur as observed by differential scanning calorimetry. FTIR did not show any significant changes of the polymer chemical structure, i.e., degradation;
5. Long-term behavior predictions of conditioned GFRP bars were made by using a method based on Arrhenius relation. These predictions provide information about the long-term tensile strength retention. To be able to use Arrhenius relation, we suppose that the mechanisms of degradation remains the same during the service life of the bars, but that they are accelerated by aging; and
6. According to the long-term predictions, the tensile strength retention of GFRP bars embedded in moist concrete will decreased by 20 and 25% after 100 and 200 years, respectively. It was shown that the service lifetime allowed to reach tensile strength retention of less than 50% should be infinite. These results are in accordance with the findings of the

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analyses reported by Mufti et al. (2007), who have tested few structures reinforced with GFRP after up to 9 years in service conditions.

Finally, this study clearly shows that the durability of tested mortar-wrapped bars, exposed to tap water, is less affected than when exposed to simulated pore-water solution. It is reasonable to assume that the latter is too harsh in comparison to the real concrete environment. The adverse consequence of such is the underestimation of GFRP durability and conservative design of concrete structures reinforced by composite materials.

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